

Reproduction Results for Learning without Forgetting

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1 Reproduction Results

This section summarizes the results obtained during the reproduction of the *Learning without Forgetting* (LwF) paper.

Two experiments were reproduced:

- ImageNet $\rightarrow$ CUB
- ImageNet $\rightarrow$ Scenes

Following the presentation style of the original paper, the absolute performance is reported for LwF, while the competing methods are presented as performance differences relative to the LwF baseline.

1.1 ImageNet $\rightarrow$ CUB

Table 1: Reproduction of the ImageNet $\rightarrow$ CUB Experiment

Method	Old (ImageNet)	New (CUB)
LwF (ours)	59.49	45.41
Fine-Tuning	-6.07	+0.12
Feature Extraction	+3.40	-3.38

Following the presentation style of the original *Learning without Forgetting* paper, the first row reports the absolute performance achieved by the LwF method. The remaining rows show performance differences relative to LwF. Positive values indicate higher performance than LwF, while negative values indicate lower performance.

1.2 ImageNet $\rightarrow$ Scenes

Table 2: Reproduction of the ImageNet $\rightarrow$ Scenes Experiment

Method	Old (ImageNet)	New (Scenes)
LwF (ours)	59.82	56.27
Fine-Tuning	-11.13	-5.45
Feature Extraction	+3.07	-1.27

Following the presentation style of the original *Learning without Forgetting* paper, the first row reports the absolute performance achieved by the LwF method. The remaining rows show performance differences relative to LwF. Positive values indicate higher performance than LwF, while negative values indicate lower performance.

1.3 Evaluation Protocol

Due to computational limitations, the ImageNet old-task evaluation was performed on the first 80 validation batches, using a batch size of 64. The same ImageNet evaluation protocol was applied consistently to all methods in both experiments.

Each experiment was executed once using a fixed random seed of 42. A one-epoch warm-up stage was performed before Fine-Tuning and LwF training, followed by five training epochs for each compared method.